DATA CLEANING & DATA TRANSFORMATION

Using Excel Power Query

Practical session | Data Analytics by Durga

Learning Objectives

Understand why data cleaning is essential before analysis.

Identify duplicates, missing values, wrong data types and inconsistent text.

Use Power Query to import, clean, transform and load data.

Build repeatable transformation steps for refreshable reporting.

Prepare data for PivotTables, dashboards and analytics.

Data Cleaning vs Data Transformation

Data Cleaning	Data Transformation
Fix errors	Change data structure
Remove duplicates	Split / merge columns
Handle missing values	Group or aggregate
Correct data types	Add calculated columns
Standardize text	Pivot / unpivot

Why Clean Data Matters

Poor-quality input produces unreliable KPIs and misleading charts.

Duplicates can inflate sales and transaction counts.

Missing values can distort averages and aggregations.

Incorrect dates prevent reliable time analysis.

Inconsistent labels can split one category into several groups.

Clean data improves accuracy and decision confidence.

What is Excel Power Query?

Power Query is Excel's import and transformation tool.

It connects to Excel, CSV, folders, databases and other structured sources.

Transformations are recorded as Applied Steps.

The original source can remain unchanged.

Core workflow: Connect → Transform → Load → Refresh.

Power Query Workflow

Connect	Transform	Validate	Load	Refresh
Import data	Clean text	Check row counts	Load table	Update source
Excel/CSV	Remove duplicates	Check nulls	Data Model	Reuse steps
Folder/DB	Change types	Check totals	Report	Maintain workflow

Practice Dataset: Order Sales

Order ID	Customer	Date	Product	Qty	Price	Total
1	John Smith	1/15/2024	Widget	5	20	100
2	Jane Doe	1/16/2024	Gadget	3	30	90
3	John Smith	1/15/2024	Widget	5	20	100
4	Alice Johnson	1/17/2024	—	—	—	—
5	Bob Brown	1/18/2024	—	2	15	30
6	Jane Doe	1/19/2024	Gadget	3	30	90

Step 1 — Import Data

Excel → Data → Get Data / From Table/Range.

Select the correct source and confirm headers.

Power Query Editor opens for transformation.

Keep raw source data separate where possible.

Power Query Editor: Key Areas

Queries pane — lists queries.

Data preview — inspect the current result.

Ribbon — cleaning and transformation commands.

Applied Steps — records the transformation sequence.

Formula bar / Advanced Editor — useful for M expressions.

Cleaning 1 — Correct Data Types

Column	Recommended Type	Purpose
Order ID	Whole Number/Text	Identifier
Order Date	Date	Time analysis
Quantity	Whole Number	Units
Price	Decimal Number	Calculation
Total Sale	Decimal Number	Sales measure
Customer Name	Text	Category/text

Cleaning 2 — Missing Values

Identify whether the blank is expected or an error.

Remove rows only when business rules justify removal.

Replace categorical blanks with 'Unknown' when appropriate.

Use Fill Down/Up when values are inherited by design.

For numeric values, define a business rule before replacing with 0 or another statistic.

Cleaning 3 — Remove Duplicates

Select columns that define a unique record.

Home → Remove Rows → Remove Duplicates.

Power Query keeps the first occurrence.

Repeated customer names are not automatically duplicate transactions.

Use an appropriate business key such as Order ID.

Cleaning 4 — Standardize Text

Transform → Format → Trim removes extra spaces.

Clean removes non-printing characters.

Use Uppercase/Lowercase/Capitalize Each Word consistently.

Replace Values can correct known variations.

Example: ' sales ', 'Sales', 'SALES' → 'Sales'.

Cleaning 5 — Dates

Set dates to the Date type rather than text.

Transform → Date can extract Year, Quarter, Month and Day.

Check locale when date formats are ambiguous.

A consistent date field supports reliable trend analysis.

Cleaning 6 — Filter and Sort

Filter invalid records using stated business rules.

Sort to inspect blanks, extreme values and unexpected categories.

Filtering in Power Query becomes a repeatable step.

Avoid manual deletion from the raw source.

Transformation — Replace Values

Before	Action	After
N/A	Replace	Unknown
sales	Standardize	Sales
SALES	Trim	Sales
Electronics	Trim	Electronics
01/15/2024 text	Change Type	Date

Transformation — Split and Merge Columns

Split Column → By Delimiter can separate text into components.

Merge Columns can create reporting keys.

Use clear delimiters such as '-' or '|'.

Keep original fields when they may be needed later.

Transformation — Add a Custom Column

Example: Total Sale = Quantity × Price.

Add Column → Custom Column.

Power Query expression: [Quantity] * [Price].

Custom columns make calculations repeatable.

Transformation — Conditional Columns

Add Column → Conditional Column creates business categories.

Example: SalesAmount >= 1000 → 'High'; otherwise → 'Standard'.

Useful for customer, product and sales segmentation.

Transformation — Group By

Group By summarizes transaction-level data.

Example: Group by Product and Sum Total Sale.

Other aggregations: Count Rows, Average, Min and Max.

Use it when a summary table is required.

Transformation — Pivot and Unpivot

Operation	Purpose	Example
Pivot Column	Values become columns	Category summary
Unpivot Columns	Columns become rows	Normalize wide data
Transpose	Switch rows/columns	Restructure special cases

Merge vs Append

Merge	Append
Combines columns	Stacks rows
Needs matching key	Needs compatible columns
Like a database join	Like UNION in SQL
Orders + Customers	January + February sales

Power Query Join Types

Join	Meaning
Left Outer	All first-table rows + matches
Right Outer	All second-table rows + matches
Inner	Matching rows only
Full Outer	All rows from both
Left Anti	First table unmatched
Right Anti	Second table unmatched

Applied Steps: The Power of Repeatability

Every transformation appears under Applied Steps.

Steps execute in sequence.

Rename important steps so the logic is understandable.

Refresh applies the same workflow to new source data.

This turns one-time cleaning into a reusable ETL process.

Load the Clean Data

Home → Close & Load sends the result to Excel.

Close & Load To provides Table/Data Model options.

Use the clean output for PivotTables, PivotCharts and dashboards.

Keep raw and transformed data logically separated.

Data Quality Validation Checklist

Check	Question
Duplicates	Are unique records actually unique?
Missing values	Which blanks remain and why?
Data types	Are dates/numbers/text correct?
Categories	Are labels standardized?
Calculations	Do totals reconcile?
Row count	Did an unexpected filter remove rows?
Refresh	Will the steps work next time?

Hands-on Activity: Clean the Order Dataset

Import the Order Sales dataset into Power Query.

Identify duplicate records.

Identify missing Product, Quantity, Price and Total Sale values.

Set appropriate data types.

Remove duplicates using an appropriate key.

Create/validate Total Sale = [Quantity] * [Price].

Review Applied Steps and Close & Load.

Expected Transformation Sequence

Step	Action
1	Import Table/Range
2	Inspect row count and columns
3	Set data types
4	Standardize text
5	Handle missing values
6	Remove duplicates
7	Create/validate Total Sale
8	Review Applied Steps
9	Close & Load

Manual Excel vs Power Query

Manual Excel	Power Query
Repeated copy/paste	Recorded steps
Can overwrite source	Source can remain unchanged
Harder to reproduce	Refreshable workflow
Formula-heavy for some tasks	Strong transformation interface
Good for one-off edits	Excellent for recurring preparation

Key Takeaways

Clean before analysis — garbage in, garbage out.

Use Power Query for repeatable and auditable preparation.

Always check data types, missing values, duplicates and category consistency.

Merge combines related tables; Append stacks similar tables.

Use Applied Steps and Refresh for recurring workflows.

Clean → Transform → Validate → Load → Analyze.

Quick Check

What is the difference between Merge and Append?

Why should duplicate customer names not be removed blindly?

Which feature records each transformation?

What data type should Order Date use?

Write the custom-column logic for Total Sale.

When would you use Group By?

FROM RAW DATA TO ANALYTICS-READY DATA

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